

# From NMF to D-SPIN

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# 1 Nonnegative Matrix Factorization for transcriptomics

## 1.1 Image-related NMF

Let's take a collection of black-and-white images containing faces. Each picture contains a total on  $n$  pixels.

Each image can be flattened from 2 dimensions to 1 by collapsing it into a vector of  $n$  pixel intensities. Given  $m$  images, the result of the flattening is a matrix of positive values (pixel intensities cannot be negative)

$$Y \in \mathbb{R}_{\geq 0}^{n \times m}$$

The image database  $Y$  is regarded as an  $n \times m$  matrix  $V$ , each column of which contains  $n$  nonnegative pixel values of one of the  $m$  facial images.

The goal of **Nonnegative Matrix Factorization (NMF)** is to decompose  $Y$  in a structure like

$$Y \approx HW$$

or, more explicitly,

$$Y_{i\mu} \approx \sum_{\alpha=1}^r W_{i\alpha} H_{\alpha\mu}$$

while imposing the **nonnegativity constraint**

$$H \in \mathbb{R}_{\geq 0}^{n \times r} \quad W \in \mathbb{R}_{\geq 0}^{r \times m}$$

The columns of  $W$  are called basis images. Each column of  $H$  is called an encoding, and is in one-to-one correspondence with a face in  $Y$ . An encoding consists of the coefficients by which a face is represented with a linear combination of basis images. Conceptually, NMF is not fundamentally different from other dimensionality-reduction techniques such as PCA: its goal is to extract a set of latent “features” that can be combined to provide a faithful reconstruction of the original data. The key difference is the nonnegativity constraint, which restricts the reconstruction to additive combinations of these features.

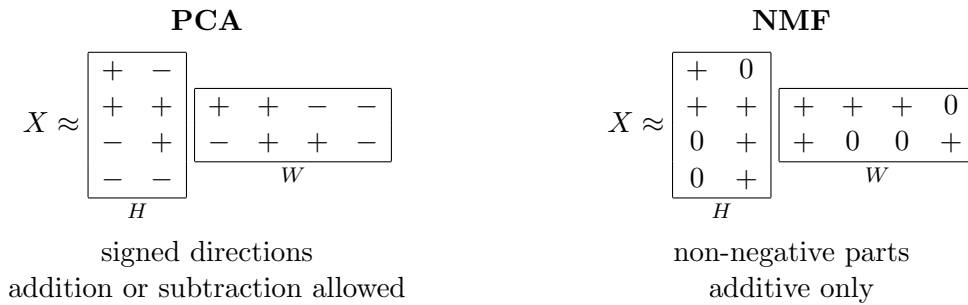


Figure 1: Comparison of PCA and non-negative matrix factorization (NMF).

The nonnegativity constraint plays a central role in determining the type of representation learned by NMF. Since both the basis images and their coefficients are nonnegative, the original image can only be reconstructed by adding the learned components together. This encourages each component to capture distinct, localized structures that can be combined to reconstruct the whole image, leading naturally to a part-based representation. Unlike methods that allow positive and negative contributions to cancel each other, NMF therefore tends to learn features corresponding to meaningful image parts.

Add citation in introduction and also figures that compare PCA to NMF.

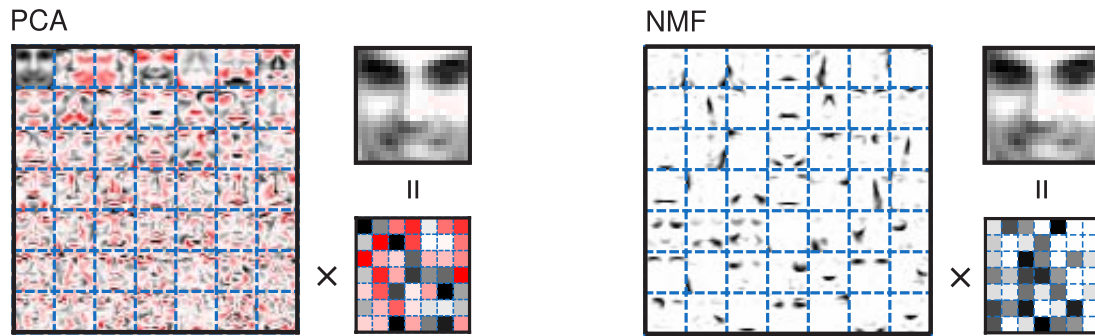


Figure 2: Image decomposition. PCA constructs a set of *eigenfaces*, while NMF favours a parts-based representation

## 1.2 Bridging to genomics

Additive-only combinations are quite interesting when it comes to genomics, as a "negative" expression of a gene is something that cannot happen in nature and it is of difficult interpretation and understanding.

Therefore, we aim to build a correspondence which looks something like this

| Image NMF                   | Transcriptomics                             |
|-----------------------------|---------------------------------------------|
| <b>source image</b>         | <b>sample / spatial location</b>            |
| <b>pixels</b>               | <b>genes</b>                                |
| <b>basis image</b>          | <b>gene-expression program / factor</b>     |
| <b>encoding of an image</b> | <b>factor activity in a sample/location</b> |

We can think of **gene program** as set or pattern of genes whose expression tends to vary together and that collectively reflects a biological process, cell state, or cellular function. In the specific context of NMF, the term usually refers more specifically to a learned factor characterized by a group of genes with high nonnegative weights, rather than to a predefined gene set.

### 1.2.1 Source images

In image decomposition, you start with an image such as

one image = one observation

Each row of your matrix is one face, i.e. a set of pixel intensities which is unique to a specific face

$$Y_i = [\text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_n].$$

In ordinary transcriptomics, we can make exactly the same construction. Suppose you have  $m$  cells and measure  $n$  genes. We go back exactly to

$$Y \in \mathbb{R}_{\geq 0}^{n \times m}.$$

but now each row is now one cell

$$Y_i = [\text{gene}_1, \text{gene}_2, \dots, \text{gene}_n].$$

The explicit analogy is

$$\boxed{\text{image} \longleftrightarrow \text{cell/sample}}$$

The image is a single observation whose features are pixel intensities.

The cell is a single observation whose features are gene-expression counts.

### 1.2.2 Pixels

images have pixel intensities

$$[\text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_n]$$

whereas a genomic sample has

$$[\text{gene}_1, \text{gene}_2, \dots, \text{gene}_n].$$

so the most direct analogy is

$$\boxed{\text{pixel} \longleftrightarrow \text{gene expression}}$$

In both cases, these are the **features** describing the observation.

There is a key difference to be noted. Pixels have a very obvious spatial relationship (e.g. pixel 100 is physically next to pixel 101), while genes generally don't have such a geometric relationship. Gene 100 is not necessarily "close" to gene 101 in any meaningful sense. When dealing with tissue samples where gene location is available, spatial information can be used. The process is an extension of NMF known as Nonnegative Spatial Factorization (NSF).

### 1.2.3 Basis images

Say that, for example, NMF discovers  $r = 10$  basis images. For  $j = 1, \dots, r$  we get

$$W_1 = [w_1^j, \dots, w_n^j]$$

Each  $W_j$  corresponds to a specific part of the faces:  $W_1$  looks like an eye,  $W_2$  looks like a nose, and so on. The key point is that a basis image is **not itself one of your original images**.

It is a learned pattern that can be used to reconstruct the original images. For one specific source image

$$\text{face} \approx 2.0(\text{eyes}) + 0.8(\text{nose}) + 1.3(\text{mouth})$$

The numbers 2.0, 0.8, 1.3 are the **encoding** for that particular face.

For transcriptomics, we replace “pixel intensity” with “gene expression”. A basis image is therefore replaced by a **gene-expression pattern across genes**.

For example, suppose NMF discovers factor 1

$$W_1 = \begin{bmatrix} 0.9, & 0.8, & 0.7, & 0.1, & 0.0, & \dots \end{bmatrix}$$

where the highly weighted genes are

$$\begin{bmatrix} \text{GAD1, GAD2, SLC6A1, COL1A1, COL1A2, } \dots \end{bmatrix}$$

which are human genes that encode core proteins for GABAergic neurotransmission, the primary chemical messaging system that slows down and calms brain activity. Then factor 1 could be interpreted as an **inhibitory-neuron-associated gene program**.

Other factors might have high weights for another set of genes corresponding to different programs.

The key analogy here is

$\text{basis image} \longleftrightarrow \text{gene-expression program}$

#### 1.2.4 Encoding

In image NMF, the encoding tells you **how much of each basis image is present in a particular image**.

Continuing with the previous example

$$\text{face}_i \approx 2.0(\text{eyes}) + 0.8(\text{nose}) + 1.3(\text{mouth})$$

The encoding is

$$H_i = (2.0, 0.8, 1.3)$$

In transcriptomics, the same thing happens. Suppose NMF finds three gene programs:

$$W_1 = \text{neuronal program}$$

$$W_2 = \text{immune program}$$

$$W_3 = \text{vascular program}$$

For cell  $i$ , NMF might give

$$H_i = (2.0, 0.1, 0.3)$$

which can be read this as

*Cell/sample  $i$  has a strong contribution from the neuronal program, very little contribution from the immune program, and some contribution from the vascular program.*

$\text{encoding} \longleftrightarrow \text{factor activity in a cell/sample}$

### 1.2.5 Final remarks

Image example:

$\text{image} \approx \text{encoding}_1 \times \text{basis image}_1 + \text{encoding}_2 \times \text{basis image}_2 + \dots$

Genomic experiment:

$\text{cell expression} \approx \text{activity}_1 \times \text{gene program}_1 + \text{activity}_2 \times \text{gene program}_2 + \dots$

Mathematically

$$Y_{i\mu} \approx \sum_{\alpha=1}^r W_{i\alpha} H_{\alpha\mu}$$

where

- $Y_{i\mu}$  = expression of gene  $\mu$  in cell  $i$ ;
- $W_{i\alpha}$  = how strongly gene  $i$  belongs to gene program  $\alpha$ ;
- $H_{\alpha\mu}$  = how active program  $\alpha$  is in cell  $\mu$ .

**One subtle difference.** NMF decomposes a face into

$$\text{face} = \text{eyes} \times c_{\text{eyes}} + \text{nose} \times c_{\text{nose}} + \text{mouth} \times c_{\text{mouth}} + \text{ears} \times c_{\text{ears}} + \dots$$

In transcriptomics,

$$\text{cell expression} = (\text{neuronal program}) \times c_{\text{neuronal}} + (\text{immune program}) \times c_{\text{immune}} + \dots$$

with different weights for each cell. The structure is the same, but we have to be careful with the analogy. **A gene program is not literally a biological “part”** in exactly the same sense as an eye is a physical part of a face. Rather, it is a **positive component of the observed expression profile**.

### 1.3 oNMF

The NMF formulation described above provides an additive representation of the data, where each observation can be reconstructed as a combination of several nonnegative programs. However, standard NMF does not prevent different programs from containing similar or overlapping genes. This can make the interpretation of the resulting programs more difficult, since several factors may describe partially overlapping biological processes.

**Orthogonal Nonnegative Matrix Factorization (oNMF)** introduces an additional constraint to address this issue. In addition to requiring all values to be nonnegative, the inferred programs are constrained to be **orthogonal**. In the context of gene-expression data, this encourages different programs to capture distinct patterns of genes rather than repeatedly describing the same transcriptional signal. The combination of nonnegativity and orthogonality therefore provides a representation that is both additive and less redundant. This is particularly useful when the objective is to identify a collection of distinct gene-expression programs.

Starting from the NMF decomposition

$$Y \approx WH$$

the nonnegativity constraints remain

$$W \geq 0 \quad H \geq 0$$

oNMF adds an orthogonality constraint to the factor representing the gene programs. Here, the concept of matrix orthogonality, which is well-defined in the context of square matrices, is used in a "broader" definition for non-square matrices

$$H^T H = \mathbb{I}_r$$

Depending on the formulation, orthogonality can be imposed on either the program matrix or the activity matrix. In the formulation used for gene-program discovery in the present context, the constraint is applied to the programs, encouraging different programs to occupy distinct directions in gene-expression space. Orthogonality can therefore be understood as an additional mechanism for separating the programs from one another.

The practical consequence can be illustrated with a simple example. Suppose that two programs are both strongly associated with the same group of neuronal genes. With ordinary NMF, there is no explicit constraint preventing both programs from containing those genes with substantial weights. The resulting factors may therefore be difficult to distinguish biologically. With oNMF, the orthogonality constraint discourages such overlap, favouring a decomposition in which the programs capture different aspects of the observed transcriptional variation.

This does not mean that two biological processes can never influence the same gene. Rather, orthogonality is a mathematical constraint on the representation learned from the data. It should therefore not be interpreted as claiming that biological pathways are themselves completely independent. Instead, it encourages the mathematical representation to assign distinct patterns of gene expression to different programs. The resulting programs can consequently be easier to interpret as separate transcriptional components.

For transcriptomics, the decomposition can therefore be understood as

$$\text{cell expression} \approx \text{activity}_1 \times \text{gene program}_1 + \text{activity}_2 \times \text{gene program}_2 + \dots$$

with two important properties. First, the contributions remain nonnegative, meaning that programs contribute additively to the reconstructed expression profile. Second, the programs are constrained to be orthogonal, reducing redundancy between them. The resulting representation can thus be viewed as a collection of distinct transcriptional patterns together with their respective activities across cells or samples.

This property is particularly relevant for perturbation experiments. Different perturbations may affect several biological processes at the same time, and their transcriptional consequences may therefore be represented by multiple programs. If these programs are sufficiently distinct, their individual activities can be examined separately, making it possible to determine which aspects of the transcriptional response are associated with each perturbation. The programs can subsequently be compared across perturbations to identify shared, opposing, or selective responses.

The interpretation of an oNMF program therefore relies on both its **gene composition** and its **activity pattern**. The genes with the highest weights provide information about the biological processes represented by the program, while the activity values indicate where or under which experimental conditions that program contributes most strongly to the observed expression profiles. A program can consequently be interpreted as a learned transcriptional pattern rather than simply as a list of genes.

This distinction is important when comparing oNMF programs with predefined biological gene sets. A predefined gene set is established independently of the dataset, for example from prior biological knowledge. An oNMF program, in contrast, is learned directly from the expression data. Its biological interpretation is obtained afterwards by examining its characteristic genes, functional enrichment, and activity across the observations. The program is therefore a **data-driven representation of coordinated gene expression**, rather than a predefined biological category.

In summary, oNMF extends the basic NMF framework by combining two constraints that are particularly useful for transcriptomic interpretation: **nonnegativity**, which supports an additive and biologically intuitive representation of gene expression, and **orthogonality**, which encourages the resulting programs to be distinct and non-redundant. This makes oNMF well suited for identifying a set of interpretable gene-expression programs and subsequently studying how those programs are activated across cells, samples, or experimental perturbations. In the context of perturbation-based single-cell analysis, these programs provide the basis for characterizing transcriptional responses and for studying the relationships between different biological processes.

## 2 Dimension-scalable Single-cell Perturbation Integration Network

This section is based on the "A guide to D-SPIN: constructing regulatory network models from single-cell RNA-seq perturbation data" paper by J. Jiang and M. Thomson, and on their prior works regarding D-SPIN.

D-SPIN (Dimension-scalable Single-cell Perturbation Integration Network) is a framework for learning probabilistic regulatory-network models from single-cell RNA-seq data collected across many perturbation conditions. Its central idea is to move from a representation of transcriptional states to a probabilistic model of how those states interact.

A natural starting point is non-negative matrix factorization (NMF), which decomposes the gene-expression matrix into latent components that can be interpreted as gene programs and their activities across cells. D-SPIN builds on this principle using orthogonal non-negative matrix factorization (oNMF) to identify positive gene programs that provide a compact representation of the expression landscape. The purpose of this factorization is therefore not yet to infer regulatory interactions, but to define a set of biologically interpretable variables on which such interactions can subsequently be modeled.

D-SPIN then takes a conceptual step beyond NMF by treating gene programs as nodes in a probabilistic graphical model. Rather than asking only which programs are active in a given cell, it asks whether the activity of one program is statistically associated with the activity of another across the population of cells. The distinction between program activity and program interaction is fundamental: the former describes the state of an individual cell, whereas the latter describes statistical dependencies between components across cells.

Perturbation conditions are incorporated into the same framework so that multiple experimental perturbations can be analyzed relative to a shared underlying interaction structure. In this formulation, perturbations act as condition-specific inputs that modify the tendency of particular programs to become active rather than requiring an entirely independent network to be inferred for every condition. This allows D-SPIN to integrate information from many perturbation experiments while maintaining a common representation of the regulatory system.

The resulting model consequently connects three levels of description: positive gene programs obtained from matrix factorization, their activity in individual cells, and the interactions that statistically couple their states. The transition from NMF to D-SPIN can therefore be explained as a transition from dimensionality reduction to probabilistic network inference: NMF provides the components with which transcriptional states are represented, whereas D-SPIN models how those components interact and respond to perturbations.

### 2.1 Basic architecture of D-SPIN

A schematic way of the principles behind D-SPIN is

$\text{genes} \longrightarrow \text{gene programs} \longrightarrow \text{program interaction network} \longrightarrow \text{perturbation response}$

More explicitly, we can think of four objects:

$W$  : gene-program composition  
 $H$  : program activities across cells  
 $J$  : interactions between programs  
 $B$  : effects of perturbations on the network

$W$  and  $H$  make up the first layer (a factorization), while  $J$  and  $B$  constitute a probabilistic model for the program states.

## 2.2 Step 1: oNMF

The first layer finds positive gene programs that can constitute a basis for the observed expression patterns. oNMF is used

$$Y \approx HW \quad Y \in \mathbb{R}_{\geq 0}^{n \times m} \quad H \in \mathbb{R}_{\geq 0}^{n \times r} \quad W \in \mathbb{R}_{\geq 0}^{r \times m}$$

with

$n$  = number of genes  
 $m$  = number of cells/samples  
 $r$  = number of gene programs

It is worth stressing that this is a modeling choice rather than a biological law. In real biology, the same gene can participate in multiple processes. Therefore, the non-overlap imposed by an orthogonal decomposition should be regarded as an interpretability constraint.

## 2.3 Step 2: From gene programs to program networks

Once the programs have been identified, each cell can be represented in a much smaller state space. Instead of  $n$  of gene-expression variables

$$G_1, G_2, \dots, G_n$$

we have a reduced number of programs  $r$

$$P_1, P_2, \dots, P_r$$

For a particular cell  $i$ , let

$$\mathbf{s}^i = (s_1^i, \dots, s_k^i), \quad i = 1, \dots, r$$

denote its program-level state (encoding).

At this point, an important conceptual change occurs. oNMF asks:

*How much of each program is present?*

D-SPIN then asks:

*Which combinations of program states are statistically favored?*

The second question is a network question.

### 2.3.1 Networks

Suppose three programs have states  $s_a$ ,  $s_b$ , and  $s_c$ . Imagine that, across a number of samples, states  $s_a$  and  $s_b$  are positively correlated

$$\mathbf{s}^1 = (s_a^1, s_b^1, s_c^1, \dots, s_k^1)$$

$$\mathbf{s}^2 = (s_a^2, s_b^2, s_c^2, \dots, s_k^2)$$

By correlation measure alone, we can not distinguish the three cases

$a \rightarrow b$  = program "a" activates program "b"

$b \rightarrow a$  = program "b" activates program "a"

$a \leftarrow c \rightarrow b$  = program "c" activates both programs "a" and "b"

Conditional-dependence model attempt to ask a more refined question

*Does the relationship between A and B remain after accounting for the other variables?*

## 2.4 Step 3. Programs interaction networks

D-SPIN uses a condition-dependent spin network, a particular type of Markov random field. The terminology comes from statistical physics.

Imagine that every gene or gene program is represented by a variable describing its state. The entire cell is then a configuration

$$\mathbf{s} = (s_1, \dots, s_K).$$

The model assigns a probability to every possible configuration. Borrowing from statistical mechanics, each  $s_i$  is the state of one microscopic degree of freedom, and the entire vector  $\mathbf{s}$  is therefore a configuration or microstate.

### 2.4.1 Spin networks in statistical mechanics

In equilibrium statistical mechanics, the probability of observing a microscopic configuration is given by the Boltzmann distribution

$$p(\mathbf{s}) = \frac{1}{Z} e^{-\beta E(\mathbf{s})}$$

where  $E(\mathbf{s})$  is the *energy of configuration*,  $\mathbf{s}$ ,  $\beta = 1/(k_B T)$  and  $Z$  is the so-called partition function

$$Z = \sum_{\mathbf{s}} e^{-\beta E(\mathbf{s})}$$

The crucial part is the definition of the energy of a single state. Without getting into details, the energy for a system of spins immersed in an external field is

$$-\beta E(\mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j$$

where  $s_i$  can only assume the two opposite values  $\pm 1$ .

The term  $h_i s_i$  describes the single-spin interaction with an external field (usually a magnetic field, when we talk about spins). Suppose  $h_i > 0$ . Then:

$$h_i(+1) = +h_i \qquad h_i(-1) = -h_i$$

Because the statistical-mechanical energy contains a minus sign, the  $s_i = +1$  state has lower energy. We can think of  $h_i$  as a local preference for the spin in the presence of an external field.

The term  $J_{ij} s_i s_j$  is a cross-interaction term between spins. The product  $s_i s_j$  tells us whether two spins are aligned:

$$s_i s_j = \begin{cases} +1, & s_i = s_j \\ -1, & s_i \neq s_j \end{cases}$$

Therefore, the sign of  $J_{ij}$  determines what the system wants the two spins to do.

If

$$J_{ij} > 0$$

then the exponent is larger when  $s_i = s_j$ . Thus, the interaction favors alignment.

If

$$J_{ij} < 0$$

then the exponent is larger when  $s_i \neq s_j$ . Thus, the interaction favors anti-alignment.

$$\boxed{J_{ij} > 0 : \text{spins want to be parallel}}$$

$$\boxed{J_{ij} < 0 : \text{spins want to be antiparallel}}$$

### 2.4.2 Program networks in transcriptomics

When moving from spin systems to transcriptomics, we recycle the energy definition

$$-\beta E(\mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j$$

but we have to be aware of what changes both in limits and interpretation.

The limits-related difference here is that spin now becomes something that describes the activity or expression level of a gene programs in a given cell, and it not limited in its values like in the statistical mechanics framework

$$\begin{array}{ccc} s_i = \pm 1 & \longrightarrow & s_i \in \mathbb{R}_+ \\ \text{spin network} & & \text{programm network} \end{array}$$

The term  $\boxed{h_i s_i}$  now relates to single-program activity tendency, while the cross-term  $\boxed{J_{ij} s_i s_j}$  says that the state of program  $i$  affects the preference for the state of program  $j$ , describing both coordinated activity between two gene programs or their antagonistic relationship.

In the transcriptomic setting, the variable ( $s_i$ ) is not a binary spin. Instead, it represents the

activity of gene program ( $i$ ) in a given cell, as inferred by oNMF in the specific case of D-SPIN. Consequently,

$$s_i \in \mathbb{R}_+$$

meaning that a program can have different degrees of activity, but its activity score is non-negative.

This difference is important when interpreting the interaction term. In the spin model,  $(s_i s_j)$  can take both positive and negative values because each spin can be either (+1) or (-1). The sign of  $(J_{ij})$  can therefore be interpreted directly in terms of whether two spins prefer to align or anti-align. This interpretation does not carry over directly to transcriptomic program activities, since

$$s_i \geq 0 \quad s_j \geq 0 \quad \implies \quad s_i s_j \geq 0 \quad \forall i, j$$

Here,  $(J_{ij})$  instead determines how the simultaneous activity of two programs contributes to the statistical preference for a cellular state. For example, if

$$J_{ij} > 0,$$

then states in which both programs have high activity receive a larger contribution to the exponent

$$J_{ij} s_i s_j,$$

and the model therefore captures a tendency for the two programs to be active together. In this sense, a positive interaction is naturally associated with co-activity or coordination between the two programs.

Conversely, if

$$J_{ij} < 0,$$

then simultaneous high activity of the two programs contributes negatively to the exponent. The model therefore penalizes states in which both programs are strongly active at the same time, capturing a statistical tendency toward mutual exclusion or antagonistic activity.

Thus, for non-negative transcriptomic program activities, it is more appropriate to interpret

$$J_{ij} > 0 : \quad \text{co-activation is statistically favored}$$

$$J_{ij} < 0 : \quad \text{simultaneous activation is statistically disfavored}$$

rather than using the language of parallel and antiparallel spins.

The magnitude  $(|J_{ij}|)$  describes the strength of this interaction within the model. Importantly, this interpretation concerns the joint statistical distribution of program activities. A positive  $(J_{ij})$  does not by itself imply that one program activates the other, just as a negative  $(J_{ij})$  does not establish that one program represses the other. The inferred interaction describes a dependency in the observed cellular states; causal interpretation requires independent biological or experimental evidence.

## 2.5 Step 4. Introducing perturbations

### 2.5.1 Spin network

Let's go back to spins and statistical mechanics. We have our spin system and we now add an external perturbation. In statistical mechanics, a particularly simple perturbation is an additional

external field acting on the spins. Suppose we apply a field  $b_i$  to spin  $i$ . The field contributes an additional term to the energy of the micro-state

$$-\beta E(\mathbf{s} | \mathbf{s}) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_i b_i s_i$$

where we left the perturbation interact with different magnitude on each spin by letting  $b_i$  depend on  $i$ .

The perturbed distribution becomes

$$p(\mathbf{s} | \mathbf{b}) = \frac{1}{Z_{\mathbf{b}}} \exp \left[ \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_i b_i s_i \right] = \frac{1}{Z_{\mathbf{b}}} \exp \left[ \sum_i (h_i + b_i) s_i + \sum_{i < j} J_{ij} s_i s_j \right]$$

where we highlighted the fact that both the energy and the probability of state  $\mathbf{s}$  are affected by the perturbation  $\mathbf{b}$ .

The perturbation does not change the interactions  $J_{ij}$ . Instead, it changes the external field experienced by the each spin

$$h_i \longrightarrow h_i + b_i$$

### 2.5.2 Transcriptomics

In transcriptomics,

$$\mathbf{s} = (s_1, \dots, s_K)$$

represents the state of the entire cell. The unperturbed model

$$p(\mathbf{s})$$

describes the distribution of cellular states under some baseline condition.

Now imagine experimentally perturbing the cell. For example, the perturbation could be a genetic knockout, activation of a transcription factor, inhibition of a signaling pathway, addition of a drug, or a change in the extracellular environment.

We therefore want to describe a conditional distribution

$$p(\mathbf{s} | q)$$

where  $q$  denotes the perturbation. The simplest way to represent perturbation  $q$  is to say that it produces an additional effective field on each gene or gene program  $i$ . We denote this field by

$$b_{iq} = \text{effect of perturbation } q \text{ on program } i$$

The perturbed distribution is consequently

$$p(\mathbf{s} | q) = \frac{1}{Z_q} \exp \left[ \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_i b_{iq} s_i \right]$$

The sign of  $b_{kq}$  determines the direct effect of perturbation  $q$  on variable  $s_k$ :

$$b_{kq} \begin{cases} > 0 & \Rightarrow \text{favors larger } s_k \\ < 0 & \Rightarrow \text{favors smaller } s_k \end{cases}$$

Remarkably, even when  $b_{iq} = 0$ , the state  $s_i$  can still respond through the pairwise couplings  $J_{ij}$ .

This, the introduction of perturbations can be summarized as

$$\boxed{\text{perturbation} \rightarrow b_{iq} \text{ (direct)} \rightarrow J_{ij} \text{ (network-mediated)} \rightarrow \text{cellular response}}$$

The parameter  $b_{iq}$  should be interpreted as the way in which perturbation  $q$  enters the model, rather than as a direct measurement of the biological effect of  $q$  on gene  $i$ . In this sense,  $b_{iq}$  represents an ineffective perturbation field: it describes the direct influence of the intervention on variable  $s_i$ . By contrast, the pairwise couplings  $J_{ij}$  describe how changes in one variable can propagate through the system. The resulting cellular response therefore emerges from the combination of the perturbation fields and the underlying interaction network.

This distinction is particularly useful when studying causality. By comparing the distribution of cellular states before and after a controlled perturbation, one can ask whether changing one component of the system produces a systematic change in another.

**Example 1.** Consider three gene programs,

$$s_1 = \text{interferon} \quad s_2 = \text{inflammation} \quad s_3 = \text{cell cycle}$$

Suppose the baseline model contains a positive coupling between the interferon and inflammation programs,

$$J_{12} > 0$$

This means that the model favors cellular configurations in which these two programs tend to be active together. Now consider a perturbation  $q$  that activates the interferon pathway. Suppose

$$b_{1q} > 0.$$

The perturbation directly pushes the interferon program toward larger values of  $s_1$ . Because

$$J_{12} > 0,$$

the increase in  $s_1$  can also make configurations with larger  $s_2$  more probable. Thus, one could observe

$$\Delta s_1 > 0$$

as a direct response and

$$\Delta s_2 > 0$$

as a network-mediated response. The model is therefore not simply saying that perturbation  $q$  changes gene program 1. Rather, it says that perturbation  $q$  changes the probability landscape of the entire cellular state through both its direct effects  $b_{iq}$  and the existing interactions  $J_{ij}$ .

**Example 2.** Consider two gene programs,  $s_A$  and  $s_B$ . In observational data, we may find that they are positively associated

$$\langle s_A s_B \rangle - \langle s_A \rangle \langle s_B \rangle > 0$$

However, this association alone does not establish that  $A$  regulates  $B$ . The two programs could, for example, be responding to a common upstream regulator.

Now suppose we experimentally perturb program  $A$ . This can be represented by a perturbation field  $b_{Aq}$ , while initially setting

$$b_{Bq} = 0$$

If the perturbation produces a systematic change in the distribution of  $s_B$ ,

$$p(s_B | q) \neq p(s_B)$$

then changing  $A$  has an effect on  $B$ . In the simplest interpretation, the causal chain is

$$A \xrightarrow{q} B$$

The pairwise model provides a mechanism for representing this response through the interaction network

$$A \xrightarrow{b_{Aq}} \text{network response} \xrightarrow{J_{AB}, J_{AC}, \dots} B$$

Thus, perturbations provide information that is not available from observational correlations alone: they actively change one part of the system and allow us to measure how the rest of the system responds. A controlled perturbation therefore helps distinguish *association* from *response to intervention*, which is the key step toward identifying causal relationships.

## 3 Introduction to the thesis use-case

### 3.1 Single-cell genetic perturbation profiling

Understanding how genes control cellular phenotypes requires methods that can connect a specific genetic perturbation to its molecular consequences. Classical genetic screens have traditionally relied on relatively simple phenotypic readouts, such as cell viability, proliferation, or the expression of a selected marker. Although these approaches are highly scalable, they reduce the response of a cell to a small number of measurements and can therefore obscure the molecular mechanisms underlying a phenotype. Single-cell RNA sequencing (scRNA-seq), in contrast, provides a high-dimensional description of the transcriptional state of individual cells. Perturb-seq combines these two approaches by coupling pooled CRISPR-based genetic perturbations with single-cell transcriptomics, thereby enabling the molecular consequences of many perturbations to be measured simultaneously [?].

#### 3.1.1 scRNA-seq

Single-cell transcriptomic profiling enables the characterization of gene-expression programs at the resolution of individual cells. Unlike bulk RNA sequencing, which averages transcript abundance across heterogeneous cell populations, single-cell RNA sequencing (scRNA-seq) captures transcriptomes from individual cells, allowing cellular heterogeneity and rare populations to be resolved. Typically, cells are isolated or partitioned into individual reaction compartments, where RNA molecules are converted into cDNA and labelled with cell-specific barcodes. Sequencing data are subsequently processed to generate a gene-by-cell expression matrix, which can be analysed using dimensionality-reduction and clustering approaches to identify transcriptionally distinct cell states or populations. Additional analyses can characterize marker genes, cellular trajectories, and changes in gene-expression programs across experimental conditions. Consequently, single-cell transcriptomics provides a high-resolution framework for investigating cellular composition, functional heterogeneity, and dynamic transcriptional responses within complex biological systems.

#### 3.1.2 CRISPR

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) is a genome-engineering technology derived from a bacterial adaptive immune system. In its commonly used form, the CRISPR-associated nuclease Cas9 is guided to a specific DNA sequence by a single-guide RNA (sgRNA), where it introduces a targeted double-strand break. Cellular DNA-repair pathways can then be exploited to disrupt genes through small insertions or deletions, a strategy known as CRISPR knockout (CRISPR-KO), or to introduce defined sequence changes. In contrast, CRISPR interference (CRISPRi) uses a catalytically inactive Cas protein (dCas9) to bind regulatory regions and sterically hinder transcription, reducing gene expression without altering the underlying DNA sequence.

CRISPRa (CRISPR activation) extends this principle to transcriptional regulation without altering the underlying DNA sequence. It uses a catalytically inactive Cas protein (dCas9) fused to, or recruited alongside, transcriptional activation domains. Guided to promoter or enhancer regions by an sgRNA, the dCas9-based complex increases transcription of the targeted gene. CRISPRa therefore enables programmable, reversible modulation of endogenous gene expression while preserving genomic sequence integrity.

### 3.1.3 Perturb-seq

Perturb-seq (Perturbation-sequencing) is a high-throughput approach that combines pooled genetic perturbations with single-cell RNA sequencing to link specific perturbations to their transcriptional consequences at single-cell resolution. In a Perturb-seq experiment, cells are exposed to a library of single-guide RNAs (sgRNAs), with each guide targeting a particular gene or regulatory element. The perturbation identity is subsequently associated with the transcriptome of the same cell. Consequently, every sequenced cell can be regarded as an individual perturbation-response measurement: the sgRNA defines the experimental intervention, while the expression profile provides a high-dimensional phenotype.

This experimental design provides an important conceptual advantage over bulk measurements. A perturbation does not necessarily produce a homogeneous response: cells carrying the same perturbation may occupy different transcriptional states because of stochasticity, cell-cycle differences, technical variation, or incomplete perturbation efficiency. These effects can be directly observed with single-cell measurements rather than being averaged away. At the same time, pooled screening substantially reduces the experimental cost compared with performing a separate RNA-seq experiment for every perturbation.

Perturb-seq is therefore particularly suited to the study of gene regulatory networks and genetic interactions. In a combinatorial experiment, cells can receive either one or multiple perturbations, making it possible to compare single-gene and multi-gene responses. The resulting data contain information not only about the effect of individual genes but also about how their effects combine. This is important because cellular phenotypes frequently arise from interactions between regulatory pathways rather than from the independent action of individual genes.

## 3.2 The Norman et al. Perturb-seq experiment

A particularly influential application of Perturb-seq was presented by Norman et al. in 2019 in the study *"Exploring genetic interaction manifolds constructed from rich single-cell phenotypes"* [?]. The authors investigated genetic interactions in human K562 cells using CRISPR activation (CRISPRa) and single-cell transcriptomic profiling.

Unlike CRISPR knockout or CRISPR interference, CRISPRa is a gain-of-function approach: rather than reducing the expression of a target gene, it activates endogenous gene expression. This distinction is important when interpreting the resulting phenotypes, since the measured response corresponds to increased activity of the targeted gene or genes.

The experimental strategy was constructed in two stages. First, Norman et al. generated a large genetic interaction map using a growth-based CRISPRa screen. Pairs of genes were co-activated and their effects on cellular fitness were compared with the effects expected from the corresponding single-gene perturbations. This allowed the authors to identify strong genetic interactions (GIs). However, a fitness measurement is intrinsically low-dimensional: two different molecular mechanisms can result in the same change in growth. Consequently, the fitness screen identified interactions without necessarily revealing why those interactions occurred.

Perturb-seq was subsequently used to provide a richer phenotype for a selected set of these interactions. Norman et al. profiled 132 gene pairs using combinatorial CRISPRa perturbations, together with the corresponding single-gene perturbations and controls. In total, the resulting experiment contained 287 perturbation conditions considered in the construction of the transcriptional genetic-interaction landscape [?].

The central conceptual contribution of the study was to regard each transcriptional phenotype as a point in a high-dimensional space. Instead of assigning a single scalar genetic-interaction score to a gene pair, the transcriptional response provides a vector containing the expression levels of many genes. The collection of these vectors was termed a *genetic interaction manifold*. Dimensionality-reduction techniques such as UMAP could then be used to visualize relationships between perturbations. Perturbations producing similar transcriptional states were positioned close to one another, allowing groups of genes and interactions to be associated with common biological processes [?].

This representation enabled several types of biological interpretation. Perturbations could be ordered according to their transcriptional similarity, allowing the authors to infer relationships between regulatory pathways. Furthermore, the rich phenotype could distinguish different mechanistic origins of genetic interactions that would appear similar in a conventional fitness-based screen. One notable result was the identification and characterization of a synergistic interaction involving CBL and CNN1, which was associated with erythroid differentiation [?].

The Norman et al. dataset therefore illustrates an important transition in functional genomics: genetic perturbations are not treated merely as labels for a binary phenotype, but as controlled interventions that probe the structure of a high-dimensional cellular state space.

### 3.3 Cutting corners

The present work uses a preprocessed version of the Norman et al. dataset available through Figshare [?]. This resource was published by Ethan Weinberger and is explicitly described as “Preprocessed CRISPRa Perturb-seq data from Norman et al., 2019”, prepared for the study introducing ContrastiveVI+ [?]. The original experimental data are available through GEO under accession GSE133344 [?]. Preprocessed data is available here at the time of thesis writing.

#### 3.3.1 Preprocessing

The Figshare object is not raw sequencing data and therefore does not require reconstruction of the complete experimental processing pipeline from sequencing reads. Instead, it provides a single-cell expression matrix together with cell-level metadata in a form suitable for computational analysis. In particular, the dataset is distributed as a processed single-cell data object and has been used in subsequent computational studies of perturbation responses [?]. Preprocessing can be summarized as follows.

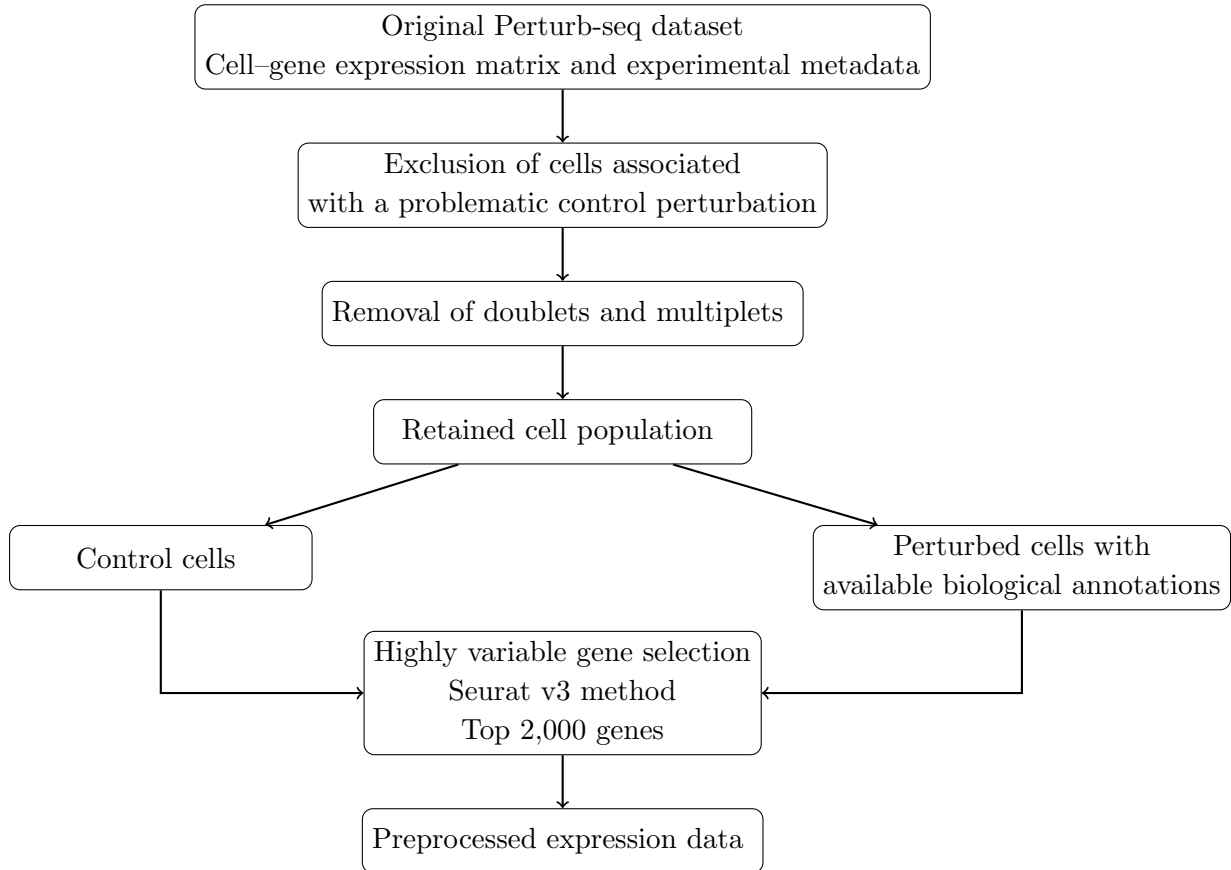
First, cells associated with a specific problematic control perturbation were excluded. Cells identified as doublets or multiplets were subsequently removed using the available cell-level metadata. Control-guide cells were retained, whereas non-control cells were retained when their perturbations had an associated gene-program annotation provided by Norman et al. This procedure restricted the analysis to cells belonging either to the control population or to perturbations with defined biological effects.

The retained perturbations were associated with several transcriptional programs described in the original study, including erythroid, granulocyte/apoptosis, pioneer-factor, pro-growth, megakaryocyte, and G1 cell-cycle-arrest responses. Thus, the resulting dataset represents a biologically annotated subset of the original experiment rather than the complete Norman et al. dataset.

After cell-level filtering, highly variable genes were identified using Scanpy’s highly-variable-gene procedure with the Seurat v3 method. Weinberger et al. selected the top 2,000 highly variable

genes for their analysis. The resulting expression representation was then used as input to ContrastiveVI+, which separates variation shared across experimental conditions from variation associated with perturbation-specific responses.

Overall, the preprocessing combines quality control, perturbation-based selection, biological annotation, and feature selection. These steps reduce technical and biological heterogeneity that is not directly relevant to the perturbation analysis while preserving control cells and annotated perturbation responses required for modelling perturbation-associated transcriptional variation.



The processed dataset is particularly convenient for downstream single-cell analysis because the perturbation identity is represented in the cell metadata. This allows cells to be grouped according to their perturbation condition and compared with control cells. For example, cells carrying a non-targeting control can be used to characterize the baseline distribution of transcriptional states, whereas cells carrying a particular CRISPRa perturbation can be treated as samples from the corresponding perturbed distribution.

An important consideration is that the presence of an sgRNA does not necessarily imply that every cell has undergone an equally strong functional perturbation. Variable guide efficiency can generate cells that contain a perturbation label but exhibit little or none of the expected transcriptional response. This issue is particularly relevant when interpreting cell-level heterogeneity. Indeed, Weinberger et al. developed ContrastiveVI+ partly because pooled CRISPR screens contain noisy perturbation labels and because some guide-positive cells can escape the functional consequences of the perturbation [?]. Consequently, perturbation identity should be interpreted as an experimental assignment rather than as a perfect measurement of perturbation strength.

### 3.4 Application of D-SPIN to the Norman dataset

The preprocessed Norman et al. dataset is particularly suitable for D-SPIN because it contains many distinct perturbation conditions measured at single-cell resolution. A D-SPIN analysis can therefore use the perturbation identity in the cell metadata as the condition variable while using the expression matrix as the molecular phenotype.

A practical analysis plan consists of several conceptual stages.

First, cells and genes should be quality-controlled and the expression matrix transformed into a representation appropriate for D-SPIN. The D-SPIN implementation expects log-normalized single-cell expression data and condition labels identifying the perturbation associated with each cell. Matched control cells are preferable because they allow perturbation responses to be estimated relative to the appropriate baseline [?].

Second, the expression space can either be represented directly at the gene level or reduced to gene programs. D-SPIN provides an unsupervised orthogonal non-negative matrix factorization (oNMF) procedure for identifying such programs, although predefined biologically motivated gene sets can also be used [?].

Third, D-SPIN jointly infers the interaction network and the perturbation-response vectors. The resulting network can then be used to determine which genes or programs interact, which perturbations act on particular parts of the network, and how different perturbations produce distinct cellular states. In the context of Norman et al., this provides a complementary perspective to the original genetic-interaction manifold: whereas Norman et al. primarily used transcriptomic similarity to organize perturbations, D-SPIN attempts to explain those observed responses through a shared network model.

This distinction also defines an important methodological advantage. Two perturbations can generate similar transcriptional profiles without acting on the same regulatory components, while two perturbations with apparently different profiles may affect different nodes of a common network. D-SPIN’s joint inference of network interactions and perturbation effects is designed to capture these relationships. The method has specifically been demonstrated on large Perturb-seq datasets, including K562 perturbation data, where it was used to investigate regulatory organization and cell-fate responses [?].

Thus, the Norman et al. dataset and D-SPIN form a natural methodological combination. Perturb-seq provides the experimentally controlled, high-dimensional perturbation-response measurements; the preprocessed Figshare object provides these measurements in a computationally accessible form; and D-SPIN provides a probabilistic framework for integrating the different perturbation conditions into an interpretable network model. The result is a progression from perturbation to transcriptional phenotype and finally to network-level interpretation, which is the central rationale for using D-SPIN in the present work.

## 4 Dataset description

The preprocessed dataset contains single-cell gene expression measurements together with cell-level annotations describing the corresponding perturbation, experimental batch, sequencing quality, and biological response. The expression matrix comprises 27,658 cells and 1,333 retained genes, with expression values represented as floating-point measurements. The matrix is sparse, with a large proportion of zero-valued entries, reflecting the characteristic sparsity of single-cell transcriptomic data.

The perturbation metadata identify 81 distinct guide conditions, including control and targeted perturbations. The experiment contains both single-target and combinatorial perturbations, resulting in a heterogeneous set of perturbation conditions with unequal numbers of cells. This imbalance is relevant for downstream modeling, since perturbations represented by fewer cells provide less information for estimating their associated transcriptional response.

Several quality-control variables are available in the metadata, including read counts, UMI counts, sequencing coverage, and indicators of coverage quality. The retained observations satisfy the available coverage criterion and correspond to individual cells rather than multiplets. These annotations therefore provide information that can be used to assess the reliability of the measurements before applying D-SPIN.

In addition to perturbation identities, the dataset contains biological program annotations assigned to cells. These annotations distinguish control cells from several transcriptional response programs associated with cellular differentiation, growth, apoptosis, and cell-cycle states. They provide an additional biological interpretation of the perturbation responses and can be useful when evaluating the programs or network structures inferred from the expression data.

The retained genes are accompanied by gene identifiers, gene names, and highly-variable-gene annotations. Consequently, the dataset provides a compact representation of the perturbation-induced transcriptional landscape while retaining the metadata required to relate molecular responses to experimental conditions. Together, these characteristics make the dataset suitable for the subsequent D-SPIN analysis, in which perturbation identities define the experimental conditions and gene expression measurements provide the observed molecular responses.

## 5 Running D-SPIN

A data analysis pipeline designed to apply D-SPIN principles to Weinberger’s dataset has been developed using Python.

The complete code, together with complete instructions set, is available on github. Code is organized in a package-like way, without being a complete python package yet. The codebase is a set of tools and functionalities to wrap and analyze results produced by the protagonist of the work, which is the `pip` installable DSPIN package [reference here to https://github.com/JialongJiang/DSPIN](https://github.com/JialongJiang/DSPIN)

### 5.1 Analysis summary

The analysis follows a stepwise workflow aimed at identifying and interpreting transcriptional programs from a perturbation-based single-cell dataset. The overall purpose is to move from the original cellular expression data toward a higher-level description of the biological programs activated or repressed by different perturbations, and to investigate how these programs relate to one another. The analysis combines model validation, identification of characteristic genes, functional interpretation, comparison of program responses, and reconstruction of a network describing relationships between programs.

Analysis steps summarized hereafter are executed in sequence by the `main.py` file of the repository.

**1. Exploratory data analysis.** The first step consists of loading the previously prepared single-cell dataset and inspecting its general structure. This provides an overview of the available expression data and cell annotations and serves as an initial check that the dataset is suitable for the subsequent analysis. Before constructing the model, the experimental annotations are also standardized. In particular, information describing the perturbation applied to each cell is organized into a consistent sample identifier, control cells are explicitly distinguished from perturbed cells, and information about experimental batches is retained. This preparation ensures that the different perturbations, control observations, and experimental groups can be correctly recognized during the analysis.

**2. Model generation.** The central part of the workflow is the construction of a model describing transcriptional programs and their relationships. Rather than considering the expression of thousands of genes independently, the analysis seeks to identify groups of genes that tend to vary together and can therefore be interpreted as coherent transcriptional programs. Each program represents a recurring pattern of gene activity that can be associated with particular cellular processes or responses. At the same time, the model captures how strongly these programs respond to the different perturbations in the dataset. The resulting representation therefore provides two complementary perspectives: what each program consists of and how each program behaves across experimental conditions.

**3. Model validation.** Once the model has been generated, its basic validity is assessed. These checks are intended to ensure that the inferred programs and their associated results are internally consistent and suitable for further interpretation. A specific validation is also performed on the underlying matrix decomposition, verifying that the identified programs and their associated components have been generated appropriately. These steps are important because the subsequent biological conclusions depend on the quality and consistency of the inferred programs.

**4. Characterization of program-associated genes.** The analysis then focuses on the genes associated with each program. For every program, the genes contributing most strongly to

its definition are identified, providing a concise representation of its molecular characteristics. Examining these highly ranked genes offers an initial indication of the biological identity of each program. For example, a group of genes involved in a common cellular process may suggest that the corresponding program represents that process. In addition to these short lists of highly characteristic genes, the analysis retains a complete ranking of genes for every program. This more comprehensive representation is subsequently used to investigate the biological functions associated with each program.

**5. Relationships between programs.** The relationships between programs are then examined by comparing their overall patterns. Programs that show similar behaviour are expected to have similar response profiles, whereas programs with opposing behaviour may represent contrasting cellular states or biological processes. A correlation analysis is therefore used to identify pairs of programs whose activities are strongly related. The strongest relationships are specifically examined to highlight potentially redundant, closely related, or biologically connected programs. This step helps determine whether the inferred programs represent clearly distinct processes or whether some of them capture related aspects of the same underlying cellular response.

**6. Functional interpretation of programs.** The biological interpretation of the programs is further developed through enrichment analysis. The complete ranked gene lists are compared with known groups of genes associated with biological functions and processes. The purpose is to determine whether the genes defining a particular program are disproportionately associated with specific biological activities. The results are subsequently summarized to provide an overview of the main functional characteristics of the programs. This allows the analysis to move beyond individual marker genes and assess the broader biological meaning of each transcriptional program.

**7. Program responses to perturbations.** A substantial part of the analysis is dedicated to understanding how the programs respond to the different perturbations. The relative activity of every program is examined across all experimental conditions, allowing perturbations that strongly activate or suppress particular programs to be identified. The strongest positive and negative responses are extracted to provide a concise overview of the most pronounced effects. This can reveal, for instance, which perturbations are associated with the strongest activation of a particular cellular process and which produce the opposite response.

**8. Reciprocal and selective responses.** The response patterns are also examined for reciprocal behaviour. This analysis investigates whether some programs tend to increase when others decrease, which may indicate opposing or complementary biological responses. In parallel, summaries of program activation are generated to characterize the overall behaviour of the programs across the perturbation landscape. The selectivity of program activation is also assessed, with the aim of determining whether a program responds broadly to many perturbations or is associated more specifically with a smaller subset of conditions. Together, these analyses provide a more detailed picture of the functional specificity of the inferred programs.

**9. Clustering of program responses.** The response patterns are subsequently visualized and compared through their correlations. Programs with similar responses across perturbations are grouped together, while programs with substantially different response patterns are separated. A hierarchical clustering analysis is used to organize the programs according to these similarities, providing an intuitive overview of groups of programs that behave in a comparable manner. Importantly, this classification is based on how the programs respond to perturbations rather than solely on the genes that constitute them. Consequently, programs that may have different molecular compositions can nevertheless be recognized as functionally related if they respond similarly to the same experimental perturbations.

**10. Directed program network.** Finally, the analysis considers the programs as components of a directed network. In this representation, each program is treated as a node, while connections between nodes describe relationships inferred from the model. Weak relationships are filtered out so that the resulting network focuses on the stronger and more meaningful connections. The direction of the relationships is retained, allowing the network to represent not only whether two programs are connected but also the inferred directionality of their association.

**11. Identification of program modules.** The network is further divided into modules, corresponding to groups of programs that are more strongly interconnected with one another. These modules provide a higher-level view of the organization of the inferred transcriptional programs and may represent groups of biological processes that participate in related cellular responses. The final network visualization therefore integrates several aspects of the preceding analysis into a single representation, showing how individual programs are connected and how they are organized into larger groups.

## 5.2 Model specifics

The DSPIN package allows both to construct single-genes networks and gene programs networks. In the case of a gene program network fit, the package performs an initial oNMF operation to create gene programs and then proceeds to build the directional network. oNMF is per-se an algorithm depending on pseudo-random number generation, thus being subject to output differences if run multiple times. Consequently, the division of genes into gene programs can natively be repeated to verify the stability of the operation starting from different random seeds. The result is a *consensus oNMF*.

The number of repetitions that generate the consensus result and the number of gene programs to split the genes into can be set by the user. At the time of writing of this thesis, no automatic detection of the optimal number gene programs is available in the DSPIN package, and developers suggest iteratively increasing the number of gene programs and interpreting results. While acknowledging that the suggested approach is the one that should ultimately be followed for publishable work, both the generation of the consensus and the fitting of the directional network are computationally demanding processes. Given that this thesis represents a first exploration of DSPIN aimed at understanding its theoretical foundations and practical applications, relatively small numbers were used:

- iterations for oNMF consensus: 25;
- gene programs: 20.

## 5.3 Consensus oNMF results

For each transcriptional program, the genes were ranked according to their weights in the corresponding oNMF component, with higher weights indicating a stronger association between a gene and the respective program. The three genes with the highest weights were selected to provide a concise representation of the main genes contributing to each inferred program. The resulting top-ranked genes are reported in Table 1, with the corresponding oNMF weights shown in parentheses. These weights are relative measures of the contribution of each gene to the respective program and were therefore used to identify the genes most strongly associated with each transcriptional component.

| Program | Gene 1                 | Gene 2            | Gene 3              |
|---------|------------------------|-------------------|---------------------|
| 1       | YBX1 (0.3064)          | ATP5I (0.2258)    | PTMA (0.1822)       |
| 2       | HIST1H4C (0.3324)      | KIAA0101 (0.2744) | TYMS (0.2180)       |
| 3       | CKS2 (0.2264)          | CENPF (0.2100)    | BIRC5 (0.2078)      |
| 4       | AP1S2 (0.2596)         | EVA1B (0.2554)    | ARHGDIB (0.2393)    |
| 5       | LST1 (0.2364)          | TYROBP (0.2166)   | CSF3R (0.2138)      |
| 6       | ENO1 (0.2095)          | SSBP1 (0.1451)    | UFD1L (0.1446)      |
| 7       | GYP A (0.3021)         | GYPB (0.3013)     | HBG2 (0.3006)       |
| 8       | UBE2L3 (0.2302)        | NDUFS5 (0.2071)   | ATP5G3 (0.2006)     |
| 9       | HBA1 (0.5772)          | HBA2 (0.4449)     | HEMGN (0.3868)      |
| 10      | SOCS2 (0.3576)         | S100A13 (0.3002)  | MAP1A (0.2026)      |
| 11      | GAL (0.3011)           | HBQ1 (0.2639)     | HK1 (0.2506)        |
| 12      | RPL14 (0.3822)         | RPS5 (0.3763)     | MT-ND2 (0.3574)     |
| 13      | CYCS (0.1853)          | SLIRP (0.1836)    | PA2G4 (0.1799)      |
| 14      | CDK1 (0.3287)          | PIF1 (0.3004)     | GTSE1 (0.2647)      |
| 15      | PLK1 (0.3106)          | CDC20 (0.3076)    | CCNB1 (0.2950)      |
| 16      | ALAS2 (0.3343)         | TSC22D1 (0.2918)  | AC006262.5 (0.2830) |
| 17      | CLC (0.4520)           | SRGN (0.4480)     | NCF1 (0.4320)       |
| 18      | CD84 (0.5835)          | COL9A2 (0.5339)   | FYB (0.3678)        |
| 19      | WDR43 (0.2623)         | NOP16 (0.2397)    | GAR1 (0.2277)       |
| 20      | RP11-301G19.1 (0.2770) | S100A11 (0.1970)  | SH3BGRL3 (0.1945)   |

Table 1: Top three genes associated with each inferred transcriptional program. Gene strengths correspond to the gene weights assigned by the oNMF model.

### 5.3.1 Enrichment

Following the characterization of the genes associated with each transcriptional program, over-representation analysis (ORA) was performed to investigate the biological processes represented within the inferred programs. For each program, the 100 highest-ranked genes were tested against the Gene Ontology (GO) Biological Process 2023 gene sets using the genes detected in the dataset as the background. GO terms with an adjusted  $p$ -value below 0.05 were considered significantly enriched. This analysis therefore provided an independent functional characterization of the transcriptional programs based on the biological processes over-represented among their constituent genes.

Several programs displayed highly coherent functional enrichments. ORA results are summarized in Table 2.

**Program 1** was significantly associated with RNA-related processes, including mRNA processing, mRNA splicing via the spliceosome, RNA splicing, and protein–RNA complex assembly. The strongest enrichment was observed for mRNA processing (adjusted  $p = 3.09 \times 10^{-5}$ ; odds ratio = 11.87), consistent with the presence of RNA-associated genes such as *YBX1*, *SNRPG*, and *SNRPE*.

**Program 2** showed a strong DNA replication and repair signature, with significant enrichment for DNA damage response, DNA metabolic process, DNA repair, DNA-templated DNA replication, and double-strand break repair via homologous recombination. DNA-templated DNA replication showed an odds ratio of 35.65, while DNA repair had an adjusted  $p$ -value of  $3.24 \times 10^{-7}$ , supporting the interpretation of this program as a DNA replication and repair-associated state.

**Program 3** exhibited a strong cell-cycle and mitotic signature. Significant terms included mitotic sister chromatid segregation, regulation of the cell cycle, positive regulation of cell-cycle

processes, and mitotic spindle organization. The enrichment for positive regulation of the cell cycle was particularly strong (odds ratio = 23.30; adjusted  $p = 3.28 \times 10^{-10}$ ). This functional profile was consistent with the presence of genes including *CENPF*, *BIRC5*, *TOP2A*, *TPX2*, and *CCNA2*.

**Program 5** showed enrichment for hematopoietic and immune-associated processes, including regulation of T-cell-mediated cytotoxicity and hemopoiesis. These findings were consistent with its expression of genes such as *LST1*, *TYROBP*, *CSF3R*, *MS4A3*, and *CEBPE*.

**Program 7** displayed a characteristic erythroid and gas-transport signature. Significant terms included oxygen transport, carbon dioxide transport, gas transport, and one-carbon compound transport, all with an adjusted  $p$ -value of approximately 0.039 and an odds ratio of 25.65. These enrichments were supported by the presence of erythroid-associated genes such as *GYPA*, *GYPB*, *HBG2*, *HBZ*, *HBG1*, and *SLC25A37*.

**Program 8** was associated with proteostasis and metabolic processes, including proteasome-mediated ubiquitin-dependent protein catabolism, ubiquitin-dependent protein catabolic processes, proteasomal protein catabolism, and cellular respiration. These enrichments were consistent with genes such as *UBE2L3*, *POMP*, *PSMA7*, *NDUFS5*, and *COX5A*.

**Program 9** showed enrichment for antigen processing and presentation of endogenous antigens, together with processes related to smooth-muscle and vascular-associated smooth-muscle proliferation. Its gene composition included *HBA1*, *HBA2*, *HEMGN*, *CNN1*, and *AHSP*, indicating a complex blood- and erythroid-associated transcriptional signature.

**Program 13** was characterized by mitochondrial processes, including protein targeting to mitochondria, insertion of proteins into the mitochondrial membrane, mitochondrial gene expression, mitochondrial organization, and translation. Protein insertion into the mitochondrial membrane showed an odds ratio of 32.39, while the enrichment of mitochondrial gene expression reached an adjusted  $p$ -value of 0.0119.

**Programs 14 and 15** were predominantly associated with cell-cycle and mitotic functions. Although Program 14 also produced an enrichment for antigen processing and presentation, its leading genes, including *CDK1*, *PIF1*, *GTSE1*, *HJURP*, and *KIFC1*, were more consistent with chromosome dynamics and cell-cycle progression. Program 15 showed significant enrichment for regulation of the mitotic metaphase/anaphase transition and positive regulation of microtubule polymerization. The latter process showed a particularly high odds ratio of 51.33 and was supported by genes including *PLK1*, *CDC20*, *CCNB1*, *CENPE*, and *DLGAP5*.

**Program 17 and Program 18** displayed immune and hematopoietic associated characteristics. Program 17 was associated with antigen processing and presentation and contained genes including *CLC*, *SRGN*, *NCF1*, *CEBPA*, and *PRG2*. Program 18 also showed enrichment for antigen processing and presentation, consistent with genes such as *CD84*, *FYB*, *ITGAM*, and *CD48*.

**Program 19** showed a particularly coherent ribosome-related signature, with significant enrichment for ribosome biogenesis, ribonucleoprotein complex biogenesis, and ribosomal large-subunit biogenesis. These findings were consistent with the presence of *WDR43*, *NOP16*, *GAR1*, *RRP1B*, and *DKC1*.

Finally, **Program 20** was significantly associated with inflammatory and host-defense processes, including positive regulation of the inflammatory response, regulation of the inflammatory response, defense response to bacterium, and positive regulation of defense response. The

strongest terms had adjusted  $p$ -values of approximately  $3.06 \times 10^{-3}$  and odds ratios of 18.49. The presence of genes such as *LAPTM5*, *VIM*, *PYCARD*, and *BTG1* further supported an immune- and stress-associated interpretation.

Not all programs produced significant GO Biological Process enrichments. In particular, programs 4, 6, 10, 11, 12, and 16 did not contain terms that passed the adjusted  $p$ -value threshold of 0.05. The absence of significant enrichment does not imply that these programs lack biological relevance; rather, their constituent genes did not form sufficiently large or specific overlaps with the tested GO categories under the applied statistical threshold. Their interpretation was therefore based primarily on their gene composition and was considered separately from the ORA results.

Overall, the enrichment analysis demonstrated that a substantial proportion of the inferred transcriptional programs corresponded to coherent biological processes. Programs associated with RNA processing, DNA replication and repair, cell-cycle progression, erythroid function, mitochondrial activity, ribosome biogenesis, and inflammatory responses were identified. These findings provided functional support for the inferred programs and demonstrated that the transcriptional decomposition captured biologically interpretable cellular processes rather than only mathematical patterns in the expression data.

| Program | Main biological processes                                       |
|---------|-----------------------------------------------------------------|
| ■ 1     | RNA processing and RNA splicing                                 |
| ■ 2     | DNA replication, DNA repair and DNA damage response             |
| ■ 3     | Cell cycle and mitotic chromosome segregation                   |
| ■ 4     | No significant GO Biological Process terms                      |
| ■ 5     | Hemopoiesis and immune/cytotoxicity-related processes           |
| ■ 6     | No significant GO Biological Process terms                      |
| ■ 7     | Oxygen, carbon dioxide and gas transport                        |
| ■ 8     | Proteasomal degradation and cellular respiration                |
| ■ 9     | Antigen processing and vascular/smooth muscle-related processes |
| ■ 10    | No significant GO Biological Process terms                      |
| ■ 11    | No significant GO Biological Process terms                      |
| ■ 12    | No significant GO Biological Process terms                      |
| ■ 13    | Mitochondrial function and mitochondrial gene expression        |
| ■ 14    | Antigen processing and presentation                             |
| ■ 15    | Mitotic progression and microtubule organization                |
| ■ 16    | No significant GO Biological Process terms                      |
| ■ 17    | Antigen processing and presentation                             |
| ■ 18    | Antigen processing and presentation                             |
| ■ 19    | Ribosome and ribonucleoprotein biogenesis                       |
| ■ 20    | Inflammatory and host-defense responses                         |

Table 2: Summary of GO Biological Process enrichment across the inferred transcriptional programs. A program was considered enriched when at least one GO term had an adjusted  $p$  – value  $< 0.05$ . Overall, 14 of 20 programs showed at least one significantly enriched GO Biological Process term.

### 5.3.2 Programs correlation

## 5.4 Relative responses to perturbations

## 5.5 Directional network

## 6 Norman from D-SPIN, and a little more